

Lavanya Mallela

B.Tech in Computer Science Engineering (2022–2026)
RGUKT Srikakulam
+91 8074163248

lavanyamallela23@gmail.com
GitHub
LinkedIn

Objective

AI/ML-focused Computer Science student with hands-on experience in machine learning, deep learning, and Large Language Models (LLMs). Built real-world applications including health modules and image classification systems, working across data preprocessing, model training, and evaluation. Seeking to apply my skills in developing scalable, production-level AI solutions.

Education

●RGUKT Srikakulam	2022–2026
B.Tech in Computer Science Engineering	CGPA: 8.7
●RGUKT Srikakulam	2020–2022
Pre-University Course	CGPA: 9.35

Experience

●AI/ML Intern, Heybobo	July 2025 – March 2026
- Developed an AI-powered Workout Plan Generator using Large Language Models (LLMs), experimenting with Qwen-7B, Qwen-14B, and other Hugging Face models. Focused on model experimentation, prompt design, and response optimization to generate personalized fitness plans based on user inputs and goals. - Designed and implemented a scalable Health Module with features including allergy management, medical hub, reminders, and vaccination scheduling using structured data pipelines. Built an OCR pipeline using PaddleOCR with Tesseract fallback for reliable medical document processing. - Built real-time pose estimation pipelines using MediaPipe (22 keypoints) for analyzing workout posture from live and recorded videos. Developed 5+ pose analysis modules and conducted extensive experimentation with 50+ pose models to improve accuracy, tracking, and performance. - Used LangGraph for multi-step workflow orchestration to manage complex AI pipelines across modules. Contributed to debugging, validation, and issue resolution of core system workflows, improving reliability and robustness.	

Projects

●PDF Query Assistant
Generative AI / NLP
- Built a RAG-based PDF Question Answering System using FAISS, and LLMs for semantic search and response generation. - Implemented multi-PDF processing with summarization and Q&A pipelines, enabling page-level retrieval and responses with source references (page numbers).
●Blood Cell Classification Using Transfer Learning
Deep Learning / Computer Vision
- Fine-tuned a ResNet50 CNN model for multi-class blood cell classification, achieving 90% accuracy. - Applied transfer learning on labeled medical image data to enable automated classification support. - Developed a GUI-based application for real-time image prediction and visualization.

Technical Skills

● Programming Languages: Python, Java, JavaScript (Basic), HTML, CSS
● AI/ML & Deep Learning: Machine Learning, Deep Learning, CNNs, Hugging Face Transformers, Model Evaluation
● Generative AI & NLP: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, LangGraph
● Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow,, MediaPipe
● Databases: MySQL, SQL
● Developer Tools & Platforms: Git, GitHub, Linux, FastAPI, Visual Studio Code, GitHub Copilot, OpenAI Codex
● Coding Platforms: LeetCode (200+ problems),HackerRank (70+ problems),GeeksforGeeks(70+ problems)

Certifications

● Deep Learning — NPTEL
● Database Management Systems — NPTEL
● Data Science with Python Workshop — Brainovision Solutions
● Microsoft Foundation Course — Microsoft
● HTML and CSS for Beginners — Microsoft